

TRONG NHAN NGUYEN

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Busan, South Korea

EDUCATION

M.S. in Biomedical Engineering Oct. 2023 – Oct. 2025
Pukyong National University, Busan 48513, Korea | GPA: 4.3/4.5
Advisor: Prof. [Junghwan Oh](#)

B.S. in Information Technology Engineering Aug. 2018 – Dec. 2022
Ho Chi Minh University of Technology, Thu Duc City, Vietnam | GPA: 3.76/4.00
Advisor: Prof. [Hung Nguyen](#)

English Proficiency: IELTS 7.5

RESEARCH INTERESTS

Image Restoration, Physics-Informed Neural Networks, Computational Imaging

ACADEMIC BACKGROUND

Research Assistant Oct. 2023 – Present
Pukyong National University, Busan, Korea
Topic: Image Restoration, Inspection Pipeline

Research Collaborator Present
University of Alabama at Birmingham, AL, USA
Collaborator: Prof. [Truong Son Hy](#)
Topic: Frequency-Space Learning for Blind Image Restoration

AI Researcher May 2022 – Nov. 2023
Artificial Intelligence Center (AIC), FPT Corporation, Vietnam
Topic: Time Series Forecasting, OCR

Undergraduate Research Assistant Feb. 2021 – Mar. 2022
Ho Chi Minh University of Technology, Vietnam

TECHNICAL SKILLS

- **Machine Learning:** Diffusion Models, Transformers, Mamba / State-Space Models, GANs, CNNs; Image Restoration (denoising, super-resolution, inpainting), Segmentation, Physics-Informed Neural Networks.
- **Frameworks & Libraries:** PyTorch, Hugging Face (Transformers, Diffusers, Accelerate), Weights & Biases, OpenCV, scikit-learn, NumPy, Pandas, Matplotlib.
- **Infrastructure & Tools:** CUDA, Multi-GPU / Distributed Training (DDP), Slurm, Linux, Git, Docker.
- **Programming Languages:** Python, C/C++, JavaScript.
- **Instrumentation & Systems:** Scanning Acoustic Microscopy (NDT), Basler cameras, Micro-Epsilon laser sensors, Yaskawa robotic arms, 3D scanners.

PUBLICATIONS

*Co-First Author, **Co-author

[Google Scholar](#)

Journal and Conference Publications

1. **Trong Nhan Nguyen**, Vu Hoang Minh Doan, Tan Hung Vo, Dang Khoa Pham, Quoc Dung Nguyen, Truong Tien Vo, Jaeyeop Choi, Junghwan Oh. “Unsupervised Hybrid Transformer–Mamba Diffusion for Depth-Dependent Image Denoising in Scanning Acoustic Microscopy,” *IEEE Transactions on Industrial Informatics*, IF = 9.8, 2026. [DOI](#)
2. **Trong Nhan Nguyen**, Vu Hoang Minh Doan, Tan Hung Vo, Jaeyeop Choi, Junghwan Oh. “Transformer-Based Super-Resolution Scanning Acoustic Imaging for Industrial Inspection,” *IEEE International Conference on Industrial Technology (ICIT 2025)*, 2025. [Link](#)
3. **Trong Nhan Nguyen**, Vu Hoang Minh Doan, Tan Hung Vo, Dinh Dat Vu, Le Hai Tran, Thi Thu Ha Vu, Truong Tien Vo, Quoc Dung Nguyen, Sudip Mondal, Jaeyeop Choi, Byeongil Lee, Junghwan Oh. “Image-enhanced Generative Models for Industrial Scanning Acoustic Super-resolution,” *Engineering Applications of*

Artificial Intelligence, IF = 9.0, 2025. [DOI](#)

4. Thi Thu Ha Vu, Tan Hung Vo, **Trong Nhan Nguyen***, Jaeyeop Choi, Le Hai Tran, Vu Hoang Minh Doan, Van Bang Nguyen, Wonjo Lee, Sudip Mondal, Junghwan Oh. "GAN-Based Super-Resolution in Linear R-SAM Imaging for Enhanced Non-Destructive Semiconductor Measurement," *Applied Sciences*, IF = 2.5, 2025. [Link](#)
5. Truong Tien Vo, Quy Phuong Le, **Trong Nhan Nguyen****, Sudip Mondal, Jaeyeop Choi, Byeongil Lee, Junghwan Oh. "A Non-Contact Ballistocardiogram-Based System with Multi-Task Deep Learning for Real-Time Vital Signs Monitoring in Acupuncture," *Computers in Biology and Medicine*, IF = 6.3, 2026. [DOI](#)
6. Thi Thu Ha Vu, Tan Hung Vo, **Trong Nhan Nguyen****, Jaeyeop Choi, Sudip Mondal, Junghwan Oh. "Optimizing Scanning Acoustic Tomography Image Segmentation With Segment Anything Model for Semiconductor Devices," *IEEE Transactions on Semiconductor Manufacturing*, IF = 2.3, 2024. [Link](#)
7. Minh Ngoc Nguyen, Khai Le-Duc, Tan-Hanh Pham, **Trong Nhan Nguyen****, Trang Nguyen, Quang Minh Luu, Ba Kien Tran, Truong-Son Hy, Viktor Dremmin, Sergei Sokolovsky, Edik Rafailov. "A Wearable Device Dataset for Mental Health Assessment Using Laser Doppler Flowmetry and Fluorescence Spectroscopy Sensors," *Communications Medicine – Nature*, IF = 7.4, 2026. [arXiv](#)

Manuscripts Under Review

1. **Trong Nhan Nguyen**, Vu Hoang Minh Doan, Tan Hung Vo, Jaeyeop Choi, Junghwan Oh. "Scanning Acoustic Image Restoration Using Diffusion for Automatic Reliable Inspection," *Computers and Electrical Engineering*, IF = 4.9, Under Review.
2. Truong Tien Vo, Huu Sang Nguyen, **Trong Nhan Nguyen****, Le Hai Tran, Quy Phuong Le, Tan Tien Nguyen, Jaeyeop Choi, Junghwan Oh. "Multimodal Smart Clothing with Haptic Feedback for Real-Time Muscle Activation Assessment in Self-Coaching Fitness," *IEEE Internet of Things*, IF = 8.9, Revision.
3. **Trong Nhan Nguyen**, Truong Son Hy, Junghwan Oh. "WaFIR: Wavelet-Fourier Mixture-of-Experts with Subband-Adaptive Routing for Blind All-in-One Image Restoration," AAAI Conference on Artificial Intelligence (AAAI), 2027 (under review).

TALKS & PRESENTATIONS

IEEE International Conference on Industrial Technology (ICIT) 2025
Presented "Transformer-Based Super-Resolution Scanning Acoustic Imaging for Industrial Inspection."

Research Seminar, Brain Korea 21 (BK21) Program 2026
Presented the annual research summary report, covering ongoing research progress in AI-based industrial inspection and image processing.

HONORS, AWARDS, AND ACHIEVEMENTS

- PKNU Global Scholarship
- Blue Scholarship, Brain Korea 21 (BK21), Korean Government Scholarship Program, 2025, 2026
- Fire Grant Scholarship, 2025, 2026
- Scholarship, Pukyong National University, 2023
- Impressive Award, HUTECH Start-up Wing, 2021
- Consolation Prize, AI Hackathon Contest, Ho Chi Minh University of Technology, 2020
- Rewards of Undergraduate Research, Ho Chi Minh University of Technology, 2020
- Outstanding Undergraduate Student Scholarship, Ho Chi Minh University of Technology, 2018

SERVICE & PROFESSIONAL DEVELOPMENT

Peer Reviewer: IEEE Sensors; IEEE International Conference on Industrial Technology; Machine Vision and Applications; Journal of Infrastructure, Policy and Development; Annals of Mathematics and Physics; Discover Artificial Intelligence; Discover Sensors.